CONTEXT AWARE SIGN LANGUAGE TRANSLATION GLOVE
PROJECT-21ECP302L

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BONAFIDE CERTIFICATE

Certified that this project report titled "**Context Aware Sign Language Translation Glove**" is the bona fide work of **VIJAY R (Reg. No.: RA2311004010071), DASS J (Reg. No.: RA2311004010072), and HARISH K (Reg. No: RA2311004010096)**, who carried out the project work under my supervision during the academic year 2025–2026 (Even Semester), as part of the course **Project (21ECP302L)** in the Department of Electronics and Communication Engineering, SRM Institute of Science and Technology, Kattankulathur.

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DECLARATION

We hereby declare that the Project entitled "**Context Aware Sign Language Translation Glove**" to be submitted for the course **"21ECP302L – Project",** is our original work as a team

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ABSTRACT

In modern healthcare systems, effective communication between patients and caregivers is extremely important. However, patients suffering from paralysis, speech impairments, or severe illness often face difficulty in expressing their needs. To address this issue, this project proposes a Smart Glove system that enables patients to communicate using simple hand gestures.

The system uses flex sensors embedded in a glove to detect finger movements. These sensor values are processed using an ESP32 microcontroller, which identifies specific gestures based on predefined thresholds and a trained machine learning model. Each gesture corresponds to a particular message such as requesting help, indicating pain, or signaling an emergency.

The processed information is transmitted to a mobile application using the Blynk IoT platform. The application displays the corresponding message and can also generate alerts during emergency situations. Additionally, a machine learning model is developed using collected sensor data to improve gesture recognition accuracy.

This system provides a low-cost, portable, and efficient solution for enhancing communication for physically challenged patients.

Sustainable Development and Goals:

SDG 3: Good Health and Well-being
→Provides quick assistance in emergency situations

SDG 10: Reduced Inequalities
→Bridges communication gaps for disabled individuals

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I would like to express my sincere gratitude to all those who have supported and guided me in the successful completion of this project titled **“**Smart Glove for Patient Assistance Using IoT and Machine Learning**.”**

I am deeply thankful to my project guide **Dr. R. Monika** for her valuable guidance, continuous support, and technical suggestions throughout the project. Her insights into gesture recognition, IoT systems, and embedded technologies were extremely helpful in completing this work successfully.

I would also like to extend my gratitude to the faculty members of the Department of Electronics and Communication Engineering for providing the necessary theoretical knowledge and resources required for this project. Their teaching in areas such as embedded systems, sensors, communication systems, and machine learning played a crucial role in building the foundation of this work.

I am thankful to my institution for providing the required facilities, development tools such as Arduino IDE**,** Blynk platform, and Google Colab, and a supportive environment to carry out this project. The availability of these tools helped in designing, developing, and testing the system effectively.

I would also like to thank my teammates for their cooperation, coordination, and contribution in various aspects of the project including hardware implementation, coding, data collection, model training, and documentation. Their teamwork made the project more efficient and successful.

Finally, I express my heartfelt thanks to my family and friends for their constant encouragement and support throughout the completion of this project.

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ABBREVIATIONS

Abbreviation	**Full Form**
IoT	Internet of Things
ML	Machine Learning
ESP32	Espressif 32-bit Microcontroller
ADC	Analog to Digital Converter
GPIO	General Purpose Input Output
WiFi	Wireless Fidelity
Blynk	IoT Mobile Application Platform
IDE	Integrated Development Environment
Vcc	Supply Voltage
GND	Ground

CHAPTER 1
INTRODUCTION

1.1 Background of the Topic

In the modern world, communication plays a crucial role in human interaction and healthcare systems. However, patients suffering from disabilities such as paralysis, stroke, or speech impairment face severe challenges in expressing their needs. These patients often rely heavily on caregivers, which may lead to delays in assistance, especially during emergency situations.

With the advancement of embedded systems, wearable devices, and Internet of Things (IoT), it has become possible to design assistive devices that can bridge this communication gap. Smart wearable systems such as sensor-based gloves provide a promising solution by capturing hand movements and converting them into meaningful outputs.

This project focuses on the development of a smart glove system that utilizes flex sensors and an Inertial Measurement Unit (IMU) to detect hand gestures. The system processes the sensor data using an ESP32 microcontroller and communicates the output to a mobile application through IoT technology.

1.2 Importance of the Project

This project is highly important in the healthcare domain as it addresses a real-world problem faced by many patients. The system provides a simple and effective communication method that reduces dependency on caregivers.

Key importance includes:

- Enables communication for speech-impaired patients
- Provides real-time alerts for emergency situations
- Reduces response time in critical conditions
- Low-cost and accessible solution
- Can be used in hospitals as well as home care environments

1.3 Problem Statement

Many patients in hospitals or homes are unable to communicate effectively due to physical limitations. Traditional communication methods such as speech or manual writing are not always practical or reliable.

Existing systems have several limitations:

- Limited gesture recognition capability

- Lack of dynamic motion detection
- Poor accuracy
- No real-time alert system

There is a need for a system that can:

- Detect multiple gestures
- Provide real-time communication
- Improve accuracy using modern technologies

1.4 Overview of the Report

This report is organized into five chapters:

- Chapter 1 introduces the background, importance, and overview of the project
- Chapter 2 discusses the literature survey and existing systems
- Chapter 3 explains system design, hardware, methodology, and implementation
- Chapter 4 presents the results and performance analysis
- Chapter 5 concludes the project and suggests future improvements

1.5 Scope of the Project

The scope of this project extends to various applications in healthcare and assistive technologies. It can be used in hospitals, rehabilitation centers, and home care environments.

Applications:

- Patient assistance systems
- Rehabilitation monitoring
- Sign language translation
- Smart healthcare systems

Limitations:

- Limited gestures
- Requires calibration
- Depends on internet connectivity

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Gesture recognition systems have gained significant attention in recent years due to their wide range of applications in healthcare, human–computer interaction, and assistive technologies. In particular, smart gloves have emerged as an effective solution for enabling communication for patients with speech or motor impairments. Various research works have explored the use of different sensors, microcontrollers, and machine learning techniques to improve the accuracy and reliability of gesture-based systems. This chapter reviews the existing literature related to flex sensor-based systems, IMU-based systems, combined sensor approaches, ESP32-based implementations, and machine learning techniques used in gesture recognition.

2.2 Flex Sensor-Based Gesture Recognition Systems

Flex sensors are widely used in smart glove systems for detecting finger bending. These sensors operate based on the principle that their resistance changes when they are bent. In many research studies, flex sensors are integrated into wearable gloves to capture finger movements and convert them into electrical signals. These signals are further processed to identify predefined gestures.

Flex sensor-based systems are advantageous due to their low cost, simple design, and ease of integration with microcontrollers. They are particularly effective for detecting static gestures such as finger bending patterns. However, these systems face limitations in detecting dynamic hand movements or orientation changes. Since flex sensors only provide information about finger bending, they cannot capture the complete motion of the hand. As a result, the accuracy of gesture recognition is limited when only flex sensors are used.

2.3 IMU-Based Gesture Recognition Systems

Inertial Measurement Units (IMUs), such as the MPU6050, are widely used in gesture recognition systems to detect motion and orientation. IMUs typically consist of an accelerometer and a gyroscope, which measure linear acceleration and angular velocity respectively. These sensors provide detailed information about hand movement, rotation, and orientation in three-dimensional space.

Research studies have shown that IMU-based systems are highly effective in detecting dynamic gestures. They are capable of capturing complex hand movements, making them suitable for applications such as rehabilitation monitoring and motion tracking. However, IMU sensors are prone to noise and drift errors, which can affect accuracy over time. Additionally, they require proper calibration and filtering techniques to ensure reliable performance.

2.4 Combined Flex Sensor and IMU-Based Systems

To overcome the limitations of individual sensors, recent research has focused on combining flex sensors with IMU sensors in smart glove systems. This hybrid approach provides a more comprehensive understanding of hand gestures by capturing both finger bending and hand motion.

In combined systems, flex sensors are used to detect the position of fingers, while the IMU captures the orientation and movement of the hand. This integration significantly improves the accuracy and robustness of gesture recognition. Studies have reported that such systems can achieve higher accuracy compared to systems that use only a single type of sensor.

The combined approach also enables the detection of both static and dynamic gestures, making it more suitable for real-world applications. However, integrating multiple sensors increases system complexity and requires efficient data processing techniques.

2.5 ESP32-Based Smart Glove Systems

The ESP32 microcontroller has become a popular choice for implementing smart glove systems due to its advanced features. It includes built-in WiFi and Bluetooth capabilities, making it suitable for IoT-based applications. Additionally, it supports Analog-to-Digital Conversion (ADC), which is essential for reading sensor data from flex sensors.

Several research works have demonstrated the use of ESP32 in gesture recognition systems. The microcontroller processes sensor data and transmits the output to mobile applications or cloud platforms. Its low power consumption and high processing capability make it ideal for wearable devices.

However, the performance of ESP32-based systems depends on efficient programming and proper handling of sensor data. Real-time processing and communication require optimized algorithms to ensure smooth operation.

2.6 Machine Learning in Gesture Recognition

Machine learning techniques have been widely used to improve the accuracy of gesture recognition systems. Traditional systems rely on threshold-based methods, which are less flexible and often inaccurate. In contrast, machine learning models can learn patterns from data and classify gestures more effectively.

Common algorithms used in gesture recognition include K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Random Forest. These models are trained using sensor data collected from different gestures. Once trained, they can predict gestures with high accuracy.

CHAPTER 3

SYSTEM DESCRIPTION AND METHODOLOGY

3.1 System Description

The Smart Glove system is designed to detect hand gestures and convert them into meaningful messages. It consists of sensors, a processing unit, and a communication module.

3.2 Block Diagram

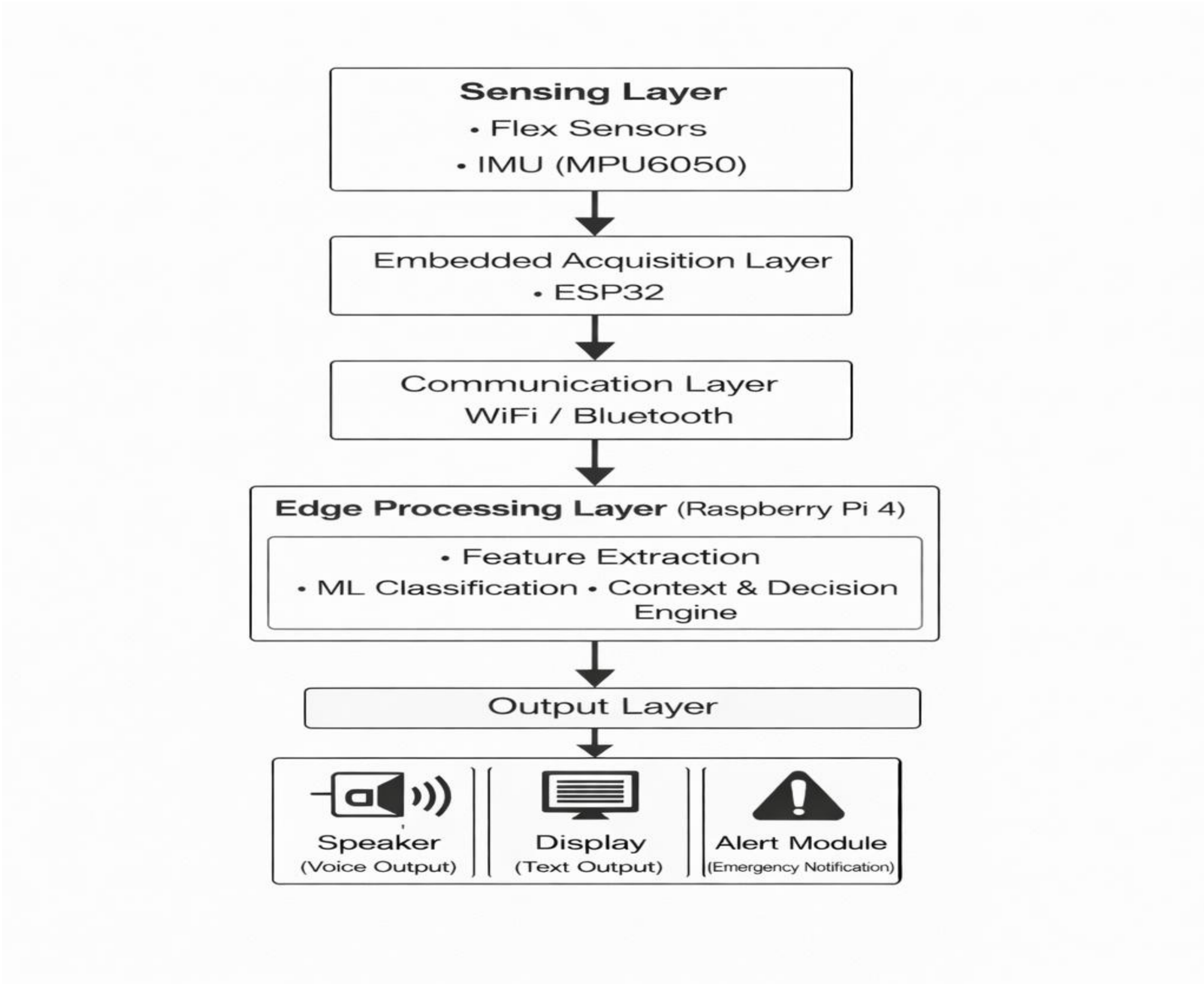

Fig 3.1: Block Diagram of Smart Glove System

3.3 Circuit Diagram

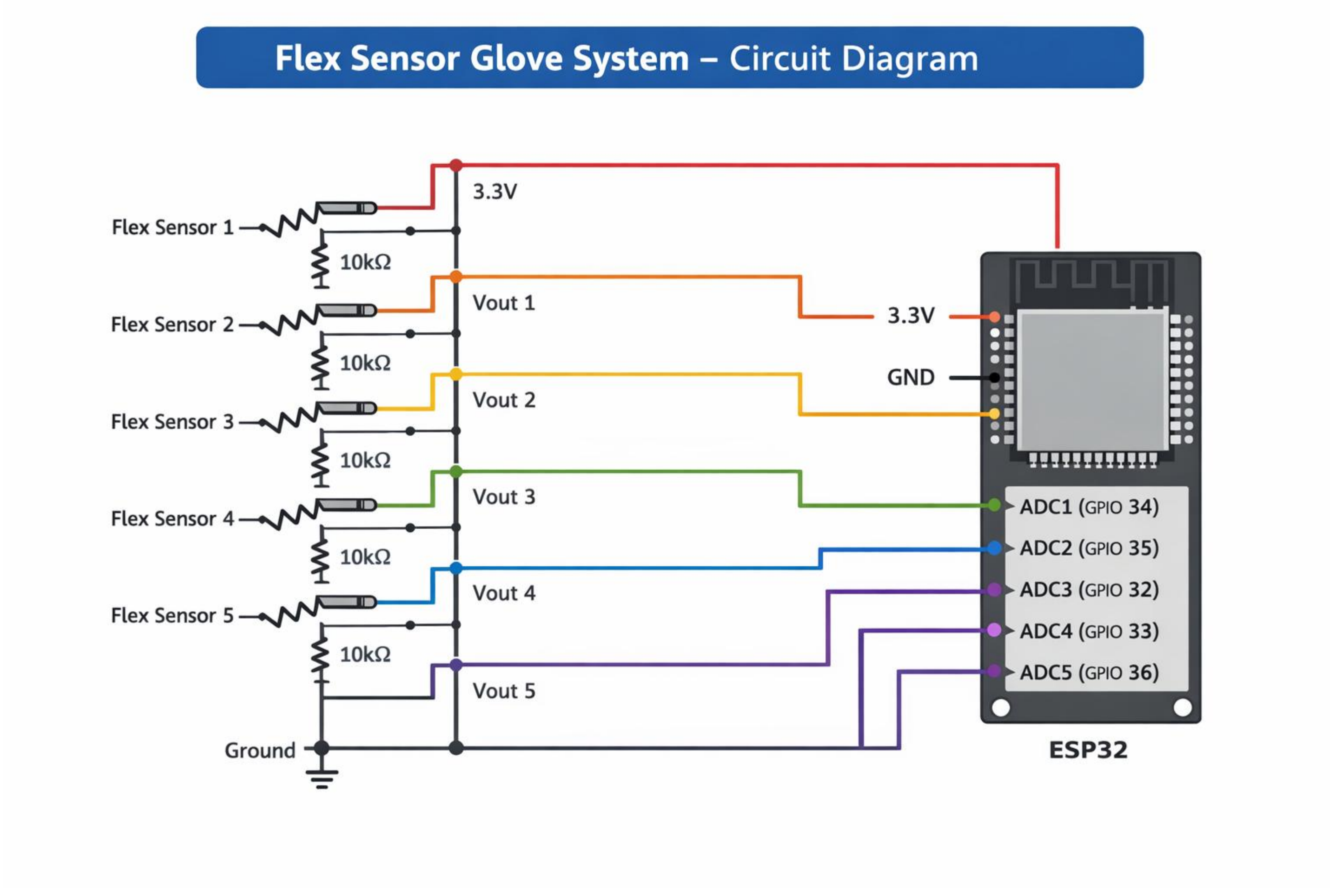

Fig 3.2 Circuit Diagram of Flex Sensor Glove using ESP32

3.3.1 Overview of the Circuit

The circuit diagram represents a Smart Glove system using flex sensors and ESP32. The purpose of the circuit is to detect finger bending and convert it into electrical signals that can be processed by the microcontroller.

Each flex sensor is connected in a voltage divider configuration, which allows the ESP32 to measure changes in resistance as voltage variations.

3.3.2 Flex Sensor Working Principle

- A flex sensor is a variable resistor.
- When the finger bends → resistance increases
- When the finger is straight → resistance is low

This change in resistance cannot be directly measured, so a voltage divider circuit is used.

3.3.3 Voltage Divider Circuit

Each flex sensor is connected with a 10kΩ resistor to form a voltage divider.

Connection:

- One end of flex sensor → 3.3V
- Other end → connected to:
    - ESP32 ADC pin
    - 10kΩ resistor → connected to GND

Working:

- When the sensor bends:
    - Resistance increases
    - Output voltage changes
- This voltage is taken as Vout

Formula:

$$\boldsymbol{V_{out} = V_{cc} \times \frac{R_{fixed}}{R_{flex} + R_{fixed}}}$$

3.3.4 ESP32 Connections

The ESP32 reads analog values from each flex sensor.

Pin Mapping: TABLE-3.1

Sensor	**ESP32 Pin**
Flex Sensor 1	GPIO 34
Flex Sensor 2	GPIO 35
Flex Sensor 3	GPIO 32
Flex Sensor 4	GPIO 33
Flex Sensor 5	GPIO 36

- These are ADC (Analog Input) pins
- ESP32 converts analog voltage → digital values

3.3.5 Power Supply

- ESP32 operates at 3.3V
- Same voltage is used for sensors
- Common ground is shared across all components

3.3.6 Signal Flow

1. User bends fingers
2. Flex sensors change resistance
3. Voltage divider converts resistance → voltage
4. ESP32 reads voltage using ADC
5. Values are processed to detect gesture
6. Message is sent to Blynk app

3.4 Hardware Components

- ESP32 Microcontroller
- Flex Sensors
- Resistors
- Connecting wires
- Power supply

3.5 Software Tools Used

- Arduino IDE for programming
- Blynk IoT platform for mobile interface
- Google Colab for machine learning model
- Python libraries (Pandas, Scikit-learn)

3.6 Methodology

Step 1: Data Collection

Sensor values are collected for different finger gestures.

Step 2: Data Processing

Collected data is organized into a dataset for training.

Step 3: Machine Learning Model

A classification model is trained to recognize gestures.

Step 4: Sensor Calibration

Threshold values are defined for bent and straight fingers.

Step 5: ESP32 Programming

ESP32 reads sensor values and identifies gestures.

Step 6: Message Mapping

Each gesture is mapped to a predefined message.

Step 7: IoT Communication

Messages are sent to Blynk app via WiFi.

3.7 Gesture Mapping

TABLE-3.2

Gesture	**Message**
All straight	I am OK
Index bent	Please come here
Middle bent	I am feeling pain
Ring bent	Please give my medicine
Index + Middle	I need assistance
All bent	Emergency!

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Results

The Smart Glove system was tested for multiple gestures, and the following results were observed:

- Flex sensors successfully detected finger movements
- ESP32 accurately processed sensor values
- Messages were displayed correctly on Blynk app
- Emergency alerts were triggered successfully

4.2 Output Screens

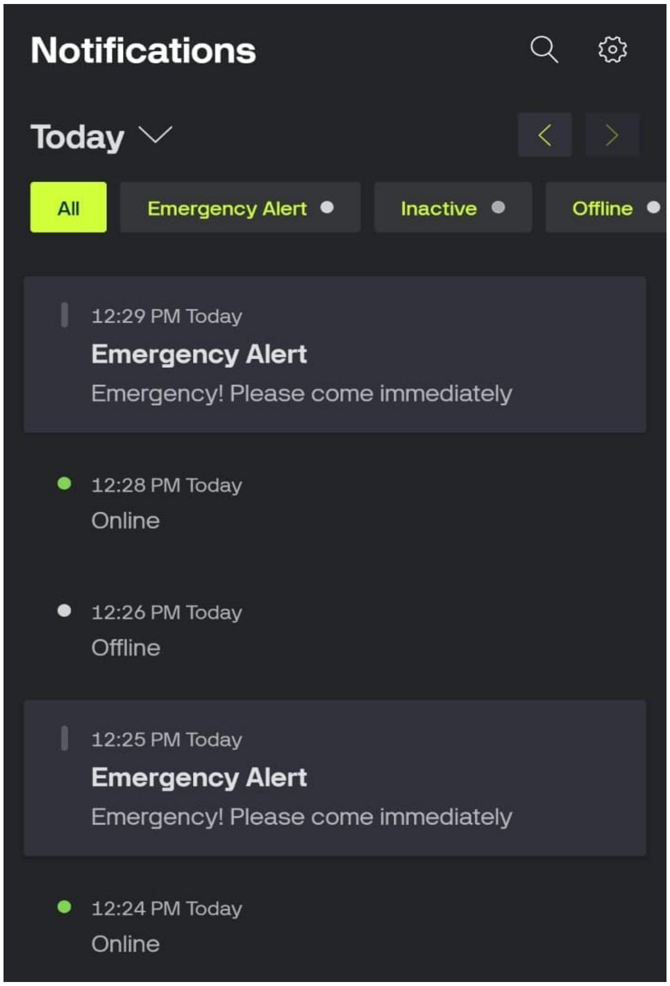

Fig 4.1 Output Display in Blynk Application

Fig 4.2 Output Display in Blynk Application

```
C:\Windows\System32\cmd.e
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
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Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Received: F1:1757 | F2:592 | Message:Emergency! Please come immediately.
Speaking: Emergency! Please come immediately.
Received: F1:876 | F2:435 | Message:Emergency! Please come immediately.
Received: F1:4095 | F2:2286 | Message:I need food.
Speaking: I need food.
Received: F1:4095 | F2:4095 | Message:Hello, I need attention.
Speaking: Hello, I need attention.
Received: F1:262 | F2:4095 | Message:Please give my medicine.
Speaking: Please give my medicine.
Received: F1:197 | F2:4095 | Message:Please give my medicine.
Received: F1:3291 | F2:2702 | Message:Emergency! Please come immediately.
```

Fig 4.3 Gesture Recognition Output Screenshot

28/04/2026, 18:14 smart glove - Colab

```
from google.colab import files
uploaded = files.upload()
```

```
Choose files smart_glove_dataset.csv
smart_glove_dataset.csv(text/csv) - 4559 bytes, last modified: 17/04/2026 - 100% done
Saving smart_glove_dataset.csv to smart_glove_dataset (1).csv
```

```
import os
print(os.listdir())
```

```
['.config', 'smart_glove_dataset.csv', 'smart_glove_dataset (1).csv', 'sample_data']
```

```
import pandas as pd

df = pd.read_csv(list(uploaded.keys())[0])
```

```
df.head()
```

	Sample_ID	F1	F2	AX	AY	AZ	GX	GY	GZ	Command
0	1	0	3569	-3292	7620	12768	-150	-985	95	HELP
1	2	0	3472	-3704	8484	13688	-1679	970	244	HELP
2	3	0	3469	-3648	7908	13664	-343	-4	62	HELP
3	4	0	3445	-3484	7932	13796	629	-6	124	HELP
4	5	0	3421	-3432	7832	13540	212	324	106	HELP

Next steps: Generate code with df New interactive sheet

```
print(df['Command'].unique())
print(df['Command'].value_counts())
```

```
['HELP' 'MEDICINE' 'HELLO' 'FOOD']
Command
HELP        25
MEDICINE    25
HELLO       25
FOOD        25
Name: count, dtype: int64
```

Fig 4.4 ML Train Model in Colab

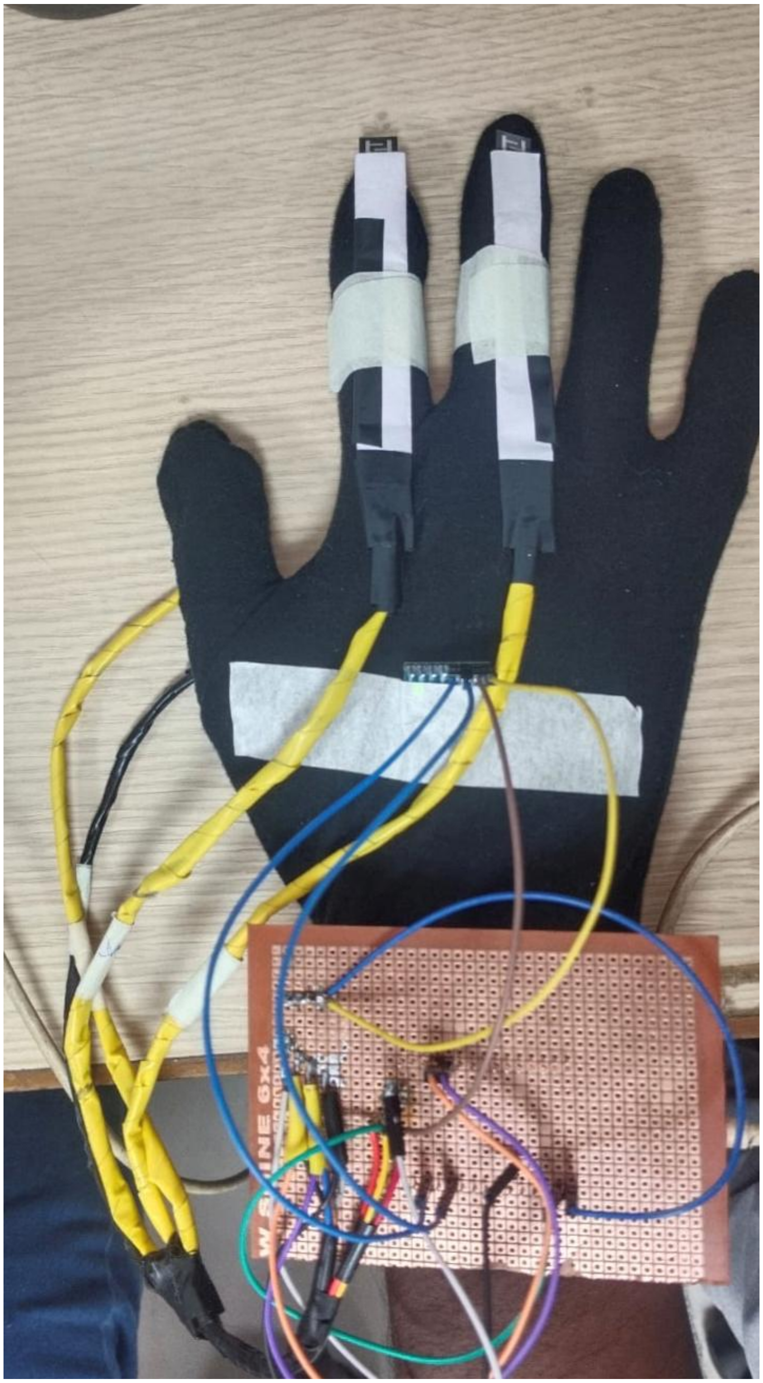

Fig 4.5 Smart Glove System Final Output

4.3 Discussion

The system performed effectively under normal conditions. The integration of machine learning improved gesture recognition accuracy compared to simple threshold-based methods.

However, sensor calibration is important to ensure consistent performance. Environmental factors and sensor quality may affect readings.

CHAPTER 5

CONCLUSION

5.1 Conclusion

The project titled "Smart Glove for Patient Assistance using Flex Sensors, IMU, ESP32, IoT, and Machine Learning" has been successfully designed and implemented to provide an effective communication system for patients with physical or speech impairments. The system utilizes flex sensors to detect finger bending and an IMU sensor to capture hand movement and orientation. These sensor inputs are processed by the ESP32 microcontroller, which identifies gestures and converts them into meaningful messages.

The integration of IoT technology enables real-time communication by transmitting the detected messages to a mobile application. This ensures that caregivers can receive patient requests instantly, improving response time and patient care. The use of both flex sensors and IMU enhances the accuracy of gesture recognition by combining static and dynamic motion detection.

Overall, the system is low-cost, portable, and easy to use, making it suitable for real-world healthcare applications. It reduces dependency on caregivers and provides a reliable method for patients to express their needs efficiently.

5.2 Future Scope

The proposed system can be further improved by incorporating advanced technologies and additional features. Machine learning models can be enhanced using deep learning techniques to improve accuracy and support a larger number of gestures. A voice output system can be added to convert gestures into speech, enabling direct communication.

The system can also be upgraded by integrating health monitoring sensors such as heart rate and temperature sensors, making it a complete smart healthcare solution. Cloud integration can be implemented to store and analyze patient data for long-term monitoring. Additionally, improvements in design, battery efficiency, and mobile application development can enhance usability and performance.

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